

AHMED MOHAMMED

Machine Learning Engineer / Applied AI / Computer Vision

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PROFESSIONAL SUMMARY

ML Engineer and M.Sc. in Artificial Intelligence student at JKU Linz (advised by Prof. Sepp Hochreiter; thesis submitted Jul 2026, degree expected 2026) with a proven track record of building and deploying ML models from research to production. Hands-on experience with PyTorch, PyTorch Lightning, diffusion models, CNNs, and LLMs across diverse data types: images (computer vision), text (NLP/LLMs), and time-series (EEG/BCI). As founder of Faultrix, architected end-to-end ML pipelines with production-grade reliability; conducted industrial ML validation at PROFACTOR GmbH funded by the Government of Upper Austria (project Zer0P). Strong fundamentals in Python, Git, Docker, and Linux.

CORE SKILLS

ML Frameworks	PyTorch • PyTorch Lightning • Scikit-learn • Diffusion Models (DDPM) • CNNs • UNet • Transformers • LLMs • Encoder-Decoder Architectures
Data Pipelines & Model Engineering	End-to-end ML pipelines • Data preprocessing • Augmentation • Structured & unstructured data
Model Engineering	Model training & optimization • Hyperparameter tuning • OOD detection • Image classification • Object detection (YOLOv8) • Inference orchestration • Structured output extraction
Evaluation & Monitoring	AUROC/FPR95 evaluation • Ablation studies • Model drift monitoring • Latency optimization • Reproducible benchmarking • Monte Carlo methods
Software & DevOps	Python (advanced) • Git • Docker • Linux • REST APIs • CI/CD concepts • Structured logging • TypeScript
Collaboration	Research-to-production handover • Technical documentation • Cross-functional teamwork • Academic writing • AI coding tools (Claude Code, Copilot)

PROFESSIONAL EXPERIENCE

Founder & ML Engineer

Jul 2025 - Present

Faultrix - Linz, Austria

Production ML system: image ingestion → model inference → structured extraction → automated reporting

- Designed and deployed an end-to-end ML pipeline handling image ingestion, model inference, structured output extraction, and automated PDF report generation – processing end-to-end in under 1 minute.
- Built fault-tolerant ML workflow with idempotent processing steps, retry logic, and iterative quality loops – achieving production-grade reliability for model inference from day one.
- Optimised LLM-powered inference components for latency and output consistency through systematic prompt engineering, structured output validation, and continuous model performance monitoring.
- Sole technical founder – owned the full ML product cycle: model selection, pipeline architecture, deployment, and iteration based on real-world performance data.

ML Researcher

Dec 2024 - Jul 2026

JKU Institute for Machine Learning - Linz, Austria

- Thesis: *Conditional Diffusion Models as Generative Classifiers for Out-of-Distribution Detection in Inkjet Print Quality Control* – supervised by Prof. Sepp Hochreiter; assistant supervisor Claus Hofmann, MSc.
- Developed a binary conditional diffusion model with a novel class-conditional separation loss,

boosting OOD detection AUROC from $92.52\% \pm 11.07\%$ to $99.03\% \pm 0.07\%$ on a CIFAR-10 airplane-vs-rest split (single ID class; +6.5 pp gain, 3-seed mean).

- Achieved 98.98% AUROC on the within-CIFAR airplane-vs-rest split (seed-42) and 90.50-96.97% across five external OOD benchmarks (CIFAR-100, Places365, FashionMNIST, SVHN, Textures).
- Built modular, reproducible experiment infrastructure using PyTorch Lightning + Hydra with configuration tracking, seed-controlled runs, and auditable raw-score artefacts.

Machine Vision Researcher

Apr 2024 - Nov 2024

PROFACTOR GmbH - Steyr, Austria (with JKU Linz)

Government of Upper Austria, project Zer0P

- Designed a multi-head conditioning mechanism and YOLO + conditional diffusion pipeline for industrial classification of inkjet experimental prints; conducted an 8-method comparison with 5-fold cross-validation on the public FTI_Zer0P benchmark (thesis-documented baseline 0.8673 ± 0.0230 AUROC).
- Evaluated production decision thresholds in a real-time inkjet quality setting (threshold-dependent operational accuracy; distinct from public AUROC baseline).

AI Research Intern

Aug 2023 - Oct 2023

Karunya University - Coimbatore, India

- Developed deep learning models for EEG-based motor imagery classification in Brain-Computer Interface (BCI) systems; designed signal preprocessing and feature extraction pipelines for multi-class EEG data.
- Evaluated and compared CNN and RNN architectures for real-time BCI decoding; documented methods and findings for academic stakeholders.

SELECTED PROJECTS

OOD Detection with Conditional Diffusion Models (M.Sc. Thesis) | JKU Institute for Machine Learning - Prof. Sepp Hochreiter
Binary CDM achieving $99.03\% \pm 0.07\%$ AUROC (3-seed mean) on CIFAR-10 airplane-vs-rest via novel separation loss. Extended to industrial inkjet quality control with multi-head conditioning (YOLOv8 + CDM two-stage pipeline). Modular implementation: PyTorch Lightning, Hydra, reproducible auditable artefacts.

Faultrix - AI-Powered Construction Report Automation | faultrix.com

Production ML SaaS: building photo → AI inference → structured damage analysis → ONORM-compliant report in <1 min. Stack: Python, Next.js, OpenAI API, Convex. Fault-tolerant pipeline with iterative quality loops.

EEG Motor Imagery Classification for Brain-Computer Interfaces | Karunya University, India

Deep learning models for multi-class EEG signal classification in BCI systems. Signal preprocessing, feature extraction pipelines, and CNN/RNN architecture comparison for real-time decoding.

EDUCATION

M.Sc. Artificial Intelligence

Oct 2024 - Jul 2026

Johannes Kepler University (JKU), Linz - Thesis submitted Jul 2026

- Thesis advisor: Prof. Sepp Hochreiter • Assistant: Claus Hofmann, MSc • Coursework: Advanced Deep Learning, Probabilistic ML, Computer Vision, Reinforcement Learning, Statistical Machine Learning

B.Sc. Mechatronics Engineering

Feb 2015 - Jan 2018

Eastern Mediterranean University, Cyprus

- Graduation project: SCARA robotic system for dynamic object tracking - real-time control and sensor integration

LANGUAGES

English	B2 – Professional Working
German	B1 – CEFR (actively improving)
Arabic	Native